Ear as an XPhone: What a Consumer EEG Headband Records of Sound Through Air, Water and Solid

Alexander Vicol and Steve Mann
University of Toronto and MannLab Canada, Toronto, Canada
vicola.lexo@gmail.com

***Abstract*—The eye as a camera reads a picture from the visual cortex. The ear as an XPhone asks the same of hearing, with X the state of matter the sound travels in: through air the ear is a microphone, through water a hydrophone, the medium of the hydraulophone, and through a solid a geophone. We report what a consumer headband (Muse S, four dry electrodes, phone app) recorded in each. *Air:* a spoken passage played 24 times; a linear decoder read its loudness envelope back from the EEG at $r = 0.36$ (shuffled null 95th percentile 0.10, $p < 0.001$; control at chance), and an EEG-conditioned music generator built by a co-worker produced clips whose rhythm followed the heard one more closely than with shuffled EEG in 6 of 6 examples, although its semantic score was not yet significant. A steady-state test of a 40 Hz-modulated tone gave no response above a floor of 0.04 μV. *Water:* six minutes with the ear at the water beside a hydraulophone gave clean EEG and a textbook optical pulse but no trace of the music, whose notes lie above the headband's 128 Hz Nyquist limit. *Solid:* four minutes of tapping and scratching the table under the head reached the head as 2.8 to 5.6 mg of vibration on the headband's accelerometer, yet the EEG neither followed that vibration (coherence at chance) nor showed a reliable tap-locked response. The ear is a microphone on a consumer headband; as a hydrophone and a geophone it needs a tagged stimulus, and we give the protocol.**

***Index Terms*—auditory EEG; speech envelope decoding; hydraulophone; underwater hearing; bone conduction; geophone; auditory steady-state response; wearable computing; consumer electronics; Mersivity.**

I. Introduction

Mann's physiphones are instruments named for the state of matter that makes their sound: the hydraulophone sounds in water, the pyrophone in fire, the ordinary instruments in air and in solids [1]–[3]. The ear as an XPhone turns that taxonomy on the listener. Sound reaches the ear through air, through water when the head is under or at the surface, and through a solid when the head rests on what is struck. Each path ends in the same cochlea, but by different routes: air conduction through the middle ear, and in water and through solids mostly bone conduction, with thresholds some 30 to 60 dB higher under water and a response that favours the low kilohertz [4], [5].

Our eye-as-camera work read what the visual cortex does with a flickering target from a consumer headband [6]. Here we ask what the same headband records of sound in each medium. The brain follows the loudness envelope of speech, and a linear decoder can read it back from the scalp [7], [8]; it follows a sound's amplitude modulation near 40 Hz as the auditory steady-state response (ASSR) [9], [10]; and the headband also carries an accelerometer and an optical pulse, which record what the medium does to the head itself [11], [12]. We report one recording per medium, and say for each what the headband gave and what the next recording must add.

II. Recording

All recordings used a Muse S headband: four dry EEG electrodes at TP9, AF7, AF8 and TP10 referenced at Fpz, 256 Hz, with an accelerometer and gyroscope and (Muse S Athena) an optical array [13], streamed to a phone running our app MuseLog [14], which records every channel and can forward the EEG to a laptop. One listener per medium; no participant other than the authors.

III. Air: the Ear as a Microphone

Speech envelope. The first two paragraphs of *Alice's Adventures in Wonderland* (a public-domain LibriVox reading, 31.5 s) were played 24 times from the speakers of a Meta Quest 3S worn by the listener, the first author, who looked at a dot. A ridge decoder on the five channels (the four Muse electrodes and a cup on the scalp) at lags of -125 to $+500$ ms read the 1 to 8 Hz envelope; each play was decoded by a model trained on the others, its first and last 3 s left out, and the mean of the held-out reconstructions is the brain's envelope of the clip (Fig. 1). With every electrode in contact it correlated with the true envelope at $r = 0.36$ over 23 plays (circular-shift null 95th percentile 0.10, $p < 0.001$), one play at 0.086 (null 0.022), and with the EEG shifted off the clips at chance (0.035, $p = 0.29$); a night session with the cup off gave 0.18 ($p < 0.001$; control 0.078). The signal sits on the ear electrodes and the cup (TP10 0.20, AF8 0.16, cup 0.15, TP9 0.14 alone). Research caps with 64 channels reach 0.10 to 0.19 for one play of continuous speech [8]; repetition gives the headband what it lacks in channels. The headset's speakers sit beside TP9 and TP10, so part of what comes back could be sound induced into the electrodes; the control that settles it, the same session with the headband unworn on the desk under the playing headset, was recorded twice but with too few complete plays (3 and 6) to decode, and is the first thing to repeat.

Music from the EEG. A co-worker, Ali Sher, ran the pipeline of Postolache et al. [15] on the four Muse channels: a

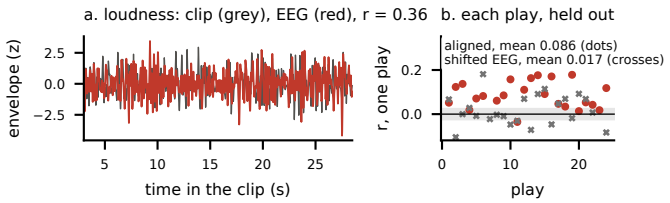


Fig. 1. The ear as a microphone. a: the loudness envelope of the 25.5 s scored from each play (grey) and the mean of the 23 held-out reconstructions from the EEG (red), $r = 0.36$. b: each play decoded by a model that never saw it, against the shift null (band) and the shifted-EEG control (crosses).

TABLE I
RHYTHM SIMILARITY OF THE SIX EXPORTED EXAMPLES TO THE HEARD CLIP (ONSET-STRENGTH CORRELATION, BEST LAG WITHIN ±60 MS). CLASSIFY-THEN-PLAY RETURNS A STORED EXAMPLE OF THE PREDICTED CLASS; FOR BOTH FAST EXAMPLES THAT FILE IS THE HEARD CLIP ITSELF.

output	fast syn.	fast str.	M2	slow syn.	slow syn. 2	slow str.	mean
EEG-conditioned	0.37	0.32	0.24	0.36	0.43	0.11	**0.31**
shuffled EEG	0.09	0.06	0.04	0.23	0.14	0.07	0.10
no EEG	0.06	0.04	0.04	0.10	0.08	0.07	0.06
classify-then-play	1.00	1.00	0.32	0.19	0.19	0.49	0.53

frozen text-to-music diffusion model [16] conditioned on the EEG through a ControlNet adapter [17], the EEG denoised with the adaptive chirplet transform [18]–[20], trained on one block of eight short rhythm stimuli and tested on the other. Scored by CLAP similarity to the true stimulus [21] over about 40 trials per block, the EEG-conditioned output came closer than a model trained on shuffled EEG–stimulus pairs on both blocks (0.496 against 0.432, 0.468 against 0.418) but not significantly, while a band-power classifier named the stimulus at twice chance on one block (25%, $p = 0.023$). We scored the six exported example clips on rhythm instead: the largest correlation, within ±60 ms, between the onset-strength curves (positive spectral flux) of the heard clip and each output (Table I). The EEG-conditioned outputs followed the heard rhythm at 0.31 on average against 0.10 with shuffled EEG and 0.06 with no EEG, higher than the shuffle in all six (sign test $p = 0.016$). Six exported examples are not a held-out test, and the conditioning carries the stimulus's rhythm class, so this says the generator draws rhythm from what it is given, a step short of reading rhythm from the brain.

A steady-state attempt. The hearing test of our exam paper, after an outline by Ali Sher, tags sound with an amplitude modulation near 40 Hz and reads the ASSR at that rate behind the ears. A 1 kHz tone modulated at 40 Hz was played from the headset in three one-minute blocks at the loudest level the listener found comfortable. With the ear electrodes at 21 and 60 μV rms and the forehead at 8, no electrode showed a 40 Hz response in any block or pooled over the three (F at most 1.4), against a floor of 0.03 to 0.04 μV; a quieter laptop run, and one with the ear electrodes off the skin, gave nothing either, and the silent baselines gave no false alarms. The response, if present at these sites, is under about 0.1 μV; the vertex, where the ASSR is largest, has no Muse electrode, and a cup there is the next step.



Fig. 2. The ear as a hydrophone: the second author at the edge of the dock with an ear at the water, wearing the headband, while the hydraulophone is played beside him.

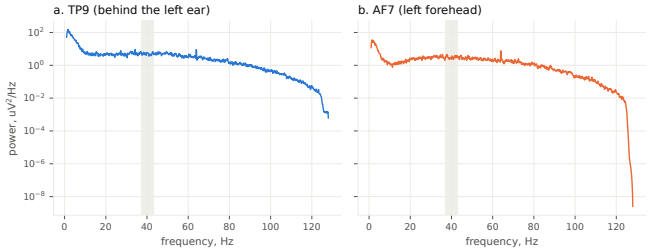


Fig. 3. Spectra of the usable epochs behind the left ear and on the left forehead at the water. The band at 37 to 43 Hz is where a 40 Hz steady-state response would appear; the line at 64 Hz is the headband's optical clock.

IV. WATER: THE EAR AS A HYDROPHONE

The second author lay at the edge of a dock in Toronto's harbour on October 4, 2026, with one ear at the water, while a hydraulophone, the water-jet instrument the second author invented, was played alongside (Fig. 2), wearing a Muse S Athena. Six minutes, no markers. The EEG was usable on 92 to 100% of 2-s epochs (under 200 μV), at 26 to 27 μV rms behind the ears and 13 to 16 on the forehead, as clean as our indoor sessions. The spectra (Fig. 3) are the ordinary 1/f shape with an alpha shoulder behind the ears; a spectrogram shows broadband bursts of movement but no narrow ridge that steps with the music, and the one steady line, at 63.9 Hz on every channel, is at the Athena's optical sampling rate and absent from recordings without the optical stream, the headband's LEDs seen by its own electrodes. A hydraulophone's fundamentals run from the low hundreds of hertz upward, all above the 128 Hz Nyquist limit, so what folds back is spread, not a line. The optical channels carried the body (Fig. 4): a pulse at 64 beats per minute on both temples, breathing near 7 per minute, and oxygenated haemoglobin rising equally on every detector pair with deoxygenated flat, the posture of a head held below the heart rather than a response to listening.

What a response would need. The floor behind the ears at 37 to 43 Hz was 4.5 μV^2/Hz. For an ASSR of rms amplitude

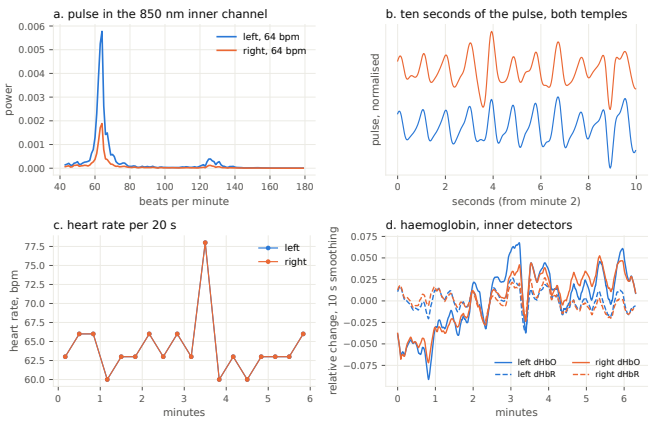


Fig. 4. The optical channels at the water. a: pulse spectrum of the 850 nm inner channel on each temple, peaking at 64 beats per minute. b: ten seconds of the pulse waveform. c: heart rate per 20 s. d: oxygenated (solid) and deoxygenated (dashed) haemoglobin on the inner detectors, relative units.

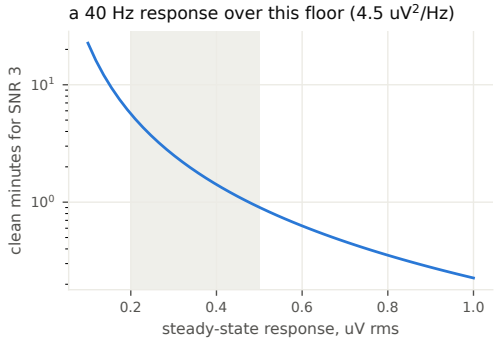


Fig. 5. Clean recording time for a 40 Hz steady-state response to reach a signal-to-noise ratio of three over the floor measured behind the ears at the water, against the response's amplitude.

A over a floor N the clean time to a signal-to-noise ratio s in one bin is $T = sN/A^2$: about one minute for 0.5 μV, three for 0.3, six for 0.2 (Fig. 5); under water the sound arrives by bone at a lower level, so the response will be at the small end.

V. Solid: the Ear as a Geophone

On November 4, 2026 the first author sat with the head resting on a table and made sound in it, recorded by MuseLog on the phone: two minutes of tapping the table with the fingernails, one of scratching it, and one of tapping it with the fingers, 4.1 minutes in all. The headband rated all four electrodes in good contact throughout (contact index 1 on every channel). Three questions follow the solid path into the head (Fig. 6).

Did the sound reach the head? Yes, mechanically. The accelerometer, high-passed at 3 Hz, shows the head vibrating at 3.8 mg rms during the fingernail taps, 2.8 during scratching and 5.6 during the finger taps, with broad spectra peaking at 6.6, 4.6 and 4.9 Hz; the table carried each gesture into the skull.

Did the EEG follow it? No. The coherence between the head's vibration and each electrode at that peak was 0.00 to 0.12, at or below each electrode's own median coherence from 1 to 15 Hz, in every segment; the 40 to 120 Hz power behind the ears was the same in all three (4.2 to 4.5 μV^2/Hz on TP9), so scratching left no broadband trace either.

Was there a response to each tap? Not reliably. Taps found as the vibration's largest peaks (22 usable epochs for the fingernails, 34 for the fingers) gave tap-locked averages behind the ears inside the band of averages at random times for the finger taps (7% of samples outside, chance 5%); the fingernail average left the band on 20% of samples, with a positive deflection of 4.5 μV near 120 ms, but it left it as far before the tap as after, so it is not a response. With taps timed only to the accelerometer's 39 Hz samples, an auditory evoked potential would also be smeared by up to 25 ms.

VI. One Protocol for Three Media

The three recordings agree on one lesson: what the headband hears of a sound is what was tagged and timed. The speech envelope came back because the same clip was played 24 times at known moments; the music, the hydraulophone and the table were not tagged, and gave the body's response (pulse, posture, vibration) rather than the brain's. The protocol that follows is the same in every medium. *A tagged source:* a speaker in air, a transducer in the water or a hydraulophone jet, a shaker on the table, each driven with an amplitude modulation at a rate the headband samples well, 40 Hz first, in 8-s on and off blocks, twenty of each, with the block edges logged on the recording's clock; then an exact permutation test over the block labels, as our visual calibration does for flicker. *Event timing for impulses:* a contact microphone or the headset's own audio clock for each tap, not the accelerometer, so an evoked potential can be averaged. *A cup at the vertex,* where auditory responses are largest and the Muse has no electrode. *A desk control* with the headband unworn beside the source, for every medium, so that induction and vibration of the electrodes are measured rather than assumed away. *The medium as the contrast:* the same tagged source heard through air, with the ear in the water, and with the head on the table, in one sitting.

Limits. One listener per medium, one session each, no markers at the water or on the table, and the air-path control not yet decodable. Nothing here is a measurement of hearing thresholds in water or through solids; it is what a consumer headband recorded, and what it would take to record more.

VII. Conclusion

Through air, the ear is a microphone on a consumer headband: a spoken passage's loudness comes back from four dry electrodes and a cup at $r = 0.36$, and an EEG-conditioned generator borrows the heard rhythm. Through water and through a solid, the headband records the body, a pulse at 64 beats per minute at the dock and the skull's vibration at the table, but not yet the brain's hearing of either. The step that made the eye a camera, a tagged stimulus with

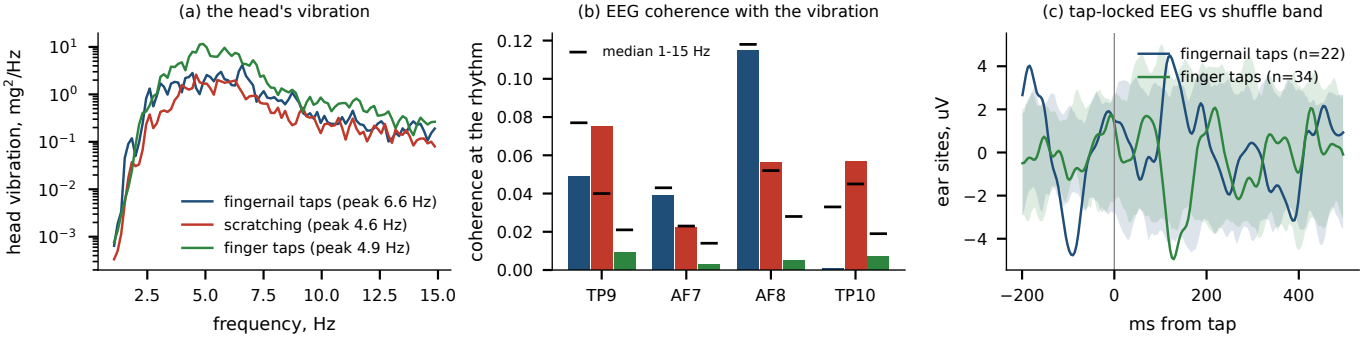


Fig. 6. The ear as a geophone: tapping and scratching the table under the head. (a) The head's vibration from the headband's accelerometer (3 Hz high-pass) in each segment. (b) Coherence between that vibration and each electrode at the segment's peak, against each electrode's median coherence from 1 to 15 Hz (black ticks). (c) EEG behind the ears averaged around the taps, against the 95% band of averages at random times in the same segment.

logged blocks and a permutation test, is the step that will make the ear a hydrophone and a geophone, and the protocol is set out above.

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References

[1] S. Mann, ""flUId Streams": Fountains that are keyboards with nozzle spray as keys that give rich tactile feedback and are more expressive and more fun than plastic keys," in *Proc. 13th ACM Int. Conf. on Multimedia*, 2005, pp. 181–190.

[2] S. Mann, R. Janzen, and M. Post, "Hydraulophone design considerations: Absement, displacement, and velocity-sensitive music keyboard in which each key is a water jet," in *Proc. 14th ACM Int. Conf. on Multimedia*, 2006, pp. 519–528.

[3] S. Mann, "Natural interfaces for musical expression: Physiphones and a physics-based organology," in *Proc. Int. Conf. on New Interfaces for Musical Expression (NIME)*, 2007, pp. 118–123.

[4] J. F. Brandt and H. Hollien, "Underwater hearing thresholds in man," *Journal of the Acoustical Society of America*, vol. 42, no. 5, pp. 966–971, 1967.

[5] H. Hollien and J. F. Brandt, "Effect of air bubbles in the external auditory meatus on underwater hearing thresholds," *Journal of the Acoustical Society of America*, vol. 46, no. 2B, pp. 384–387, 1969.

[6] S. Mann *et al.*, "The human eye as a camera," in *Proc. IEEE Int. Conf. on E-health Networking, Application and Services (HealthCom)*, 2019.

[7] J. A. O'Sullivan, A. J. Power, N. Mesgarani, S. Rajaram, J. J. Foxe, B. G. Shinn-Cunningham, M. Slaney, S. A. Shamma, and E. C. Lalor, "Attentional selection in a cocktail party environment can be decoded from single-trial EEG," *Cerebral Cortex*, vol. 25, no. 7, pp. 1697–1706, 2015.

[8] B. Accou, J. Vanthornhout, H. Van hamme, and T. Francart, "Decoding of the speech envelope from EEG using the VLAAI deep neural network," *Scientific Reports*, vol. 13, p. 812, 2023.

[9] R. Galambos, S. Makeig, and P. J. Talmachoff, "A 40-Hz auditory potential recorded from the human scalp," *Proceedings of the National Academy of Sciences*, vol. 78, no. 4, pp. 2643–2647, 1981.

[10] T. W. Picton, M. S. John, A. Dimitrijevic, and D. Purcell, "Human auditory steady-state responses," *International Journal of Audiology*, vol. 42, no. 4, pp. 177–219, 2003.

[11] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," *Physiological Measurement*, vol. 28, no. 3, pp. R1–R39, 2007.

[12] F. Scholkmann, S. Kleiser, A. J. Metz, R. Zimmermann, J. M. Pavia, U. Wolf, and M. Wolf, "A review on continuous wave functional near-infrared spectroscopy and imaging instrumentation and methodology," *NeuroImage*, vol. 85, pp. 6–27, 2014.

[13] O. E. Krigolson, C. C. Williams, A. Norton, C. D. Hassall, and F. L. Colino, "Choosing MUSE: Validation of a low-cost, portable EEG system for ERP research," *Frontiers in Neuroscience*, vol. 11, p. 109, 2017.

[14] A. Vicol and S. Mann, "MuseLog: a multi-headband acquisition app for the Muse with live OSC streaming," Software, MannLab and University of Toronto, 2026, Flutter app for Android and iOS; raw EEG, band powers, fNIRS, markers; OSC over UDP.

[15] E. Postolache, N. Polouliakh, H. Kitano, A. Connelly, E. Rodolà, L. Cosmo, and T. Akama, "Naturalistic music decoding from EEG data via latent diffusion models," *arXiv preprint arXiv:2405.09062*, 2024.

[16] H. Liu, Q. Tian, Y. Yuan, X. Liu, X. Mei, Q. Kong, Y. Wang, W. Wang, Y. Wang, and M. D. Plumbley, "AudioLDM 2: Learning holistic audio generation with self-supervised pretraining," *arXiv preprint arXiv:2308.05734*, 2023.

[17] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2023, pp. 3836–3847.

[18] S. Mann and S. Haykin, "The chirplet transform: Physical considerations," *IEEE Transactions on Signal Processing*, vol. 43, no. 11, pp. 2745–2761, 1995.

[19] N. Kumar and S. Mann, "Toward a stable and deployable adaptive chirplet transform: Residual projection, hybrid GPU acceleration, and multi-channel scalability," *arXiv preprint arXiv:2607.16629*, 2026.

[20] S. Mann, N. Kumar, J. P. Bicalho, M. Sibai, and C. Leaver-Preyra, "Adaptive chirplet transform-based sleep state detection," in *Proc. IEEE Int. Conf. on Consumer Electronics (ICCE)*, 2025, pp. 1–6.

[21] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.